APPENDIX B. SUPPLEMENTARY MAPS, FIGURES AND TABLES (ONLINE)

Table B.1: Descriptive statistics Nighttime Lights Dataset

Variable	Obs	Mean	Std. Dev.	Min	Max
Pix Region	70,581	63,250	278,774	0.740	8.731e+06
Pop Region	70,581	1.674e+06	7.237e+06	1	2.163e+08
# Region	70,581	29.69	23.76	2	89
Average nighttime light	70,581	6.630	11.97	0.01000	63

Table B.2: Descriptive statistics Gennaioli, et al. (2014) Dataset

Variable	*Obs*	*Mean*	*Std. Dev.*	*Min*	*Max*
GDP pc Country	5198	14412.16	12680.81	365.3238	62640.93
GDP pc Region	5198	13290.63	13140.64	188.9685	112293.7

Table B.3: Data sources & definitions.

Variable	**SOURCES & DEFINITIONS**
Pix Region	Definition: Area of a region measured by the number of pixel, where 1 pixel is approximately 0.86 square kilometer (at the equator). Source: GADM database of Global Administrative Areas. We use QGIS Desktop 2.4.0 with the plugin Zonal Statistics 0.1 to analyse the GIS data.
Pop Region	Definition: Regional total population. Source: Gridded Population of the World (GPW) v.3 by CIESIN.
# Region	Definition: Number of subnational regions of a country. Source: GADM database of Global Administrative Areas and Gennaioli et al. (2014).
Average nighttime light	Definition: Average nighttime light per pixel, plus 0.01 in those regions, where the average nighttime light per pixel would be zero otherwise. Source: Stable lights from National Oceanic and Atmospheric Administration (NOAA)
GDP p.c.	Definition: GDP per capita, (constant 2005 US$) based on purchasing power parity (PPP). Source: World Bank collection of development indicators and Gennaioli et al. (2014).
Area	Definition: Land area (sq. km). Land area is a country's total area, excluding area under inland water bodies, national claims to continental shelf, and exclusive economic zones. In most cases, the definition of inland water bodies includes major rivers and lakes. Source: World Bank collection of development indicators.
Population density	Definition: Population density (100,000 people per sq. km of land area). Population density is midyear population divided by land area in square kilometers. Source: World Bank collection of development indicators.
Arable land	Definition: Arable land (share of land area). Arable land includes land defined by the FAO as land under temporary crops (double-cropped areas are counted once), temporary meadows for mowing or for pasture, land under market or kitchen gardens, and land temporarily fallow. Land abandoned as a result of shifting cultivation is excluded. Source: World Bank collection of development indicators.
Transport costs	Definition: Transport costs are the product of area and pump price for gasoline in PPP (constant 2005 US$ per liter). Source: World Bank collection of development indicators.

Trade/GDP	Definition: Trade (share of GDP). Trade is the sum of exports and imports of goods and services measured as a share of gross domestic product. Source: World Bank collection of development indicators
FDI/GDP	Definition: Foreign direct investment, net inflows (share of GDP). Source: World Bank collection of development indicators.
Resource rents / GDP	Definition: Resources rents (share of GDP). Total natural resources rents are the sum of oil rents, natural gas rents, coal rents (hard and soft), mineral rents, and forest rents.. Source: World Bank collection of development indicators.
School enrollment secondary	Definition: Gross enrollment ratio is the ratio of total enrollment, regardless of age, to the population of the age group that officially corresponds to secondary education. Source: World Bank collection of development indicators.
Bureaucratic quality	Definition: Bureaucratic quality measured by the subjective index provided by the International Country Risk Guide (ICRG), where high values imply high quality of bureaucratic services. Source: The International Country Risk Guide.
Risk of internal conflict	Definition: Risk of internal conflict measured by the subjective index Internal Conflict provided by the International Country Risk Guide (ICRG). We rescale the original index so that high values imply high risk of internal conflict. Source: The International Country Risk Guide.
Polity2	Definition: Revised Combined Polity Score (Polity2), rescaled so that it ranges from -1 (strong autocracy) to +1 (strong democracy). Source: The Center for Systemic Peace (CSP) Polity IV Annual Time-Series.
Ethnic inequality	Definition: Inequality in average nighttime light between different ethnic homelands. Measures are calculated following Alesina et al. (2016). In our analysis, the inequality measures used to calculate ethnic inequality is always the same as the one used to calculate regional inequality. Hence, if we focus on the weighted Gini coefficient to measures regional inequality ethnic inequality is also measured using a weighted Gini coefficient. Source: Stable lights from National Oceanic and Atmospheric Administration (NOAA) and Geo-referencing of Ethnic Groups from Weidmann et al. (2010).

Table B.4: OLS Estimation results for model (2) with standard error adjustment for serial correlation and spatial correlation using radiuses of 1000km, 2000km and 5000km.

	(1)	(2)	(3)
VARIABLES	r:1000 km	r:2500 km	r:5000 km
log(Lighti)	0.190***	0.190***	0.190***
	(0.020)	(0.025)	(0.027)
# Observations	5,258	5,258	5,258
Region FE	YES	YES	YES
Country FE	NO	NO	NO
Country Group FE	No	No	No
Satellite FE	YES	YES	YES

Standard errors in parentheses *** p<0.01, ** p<0.05, *p<0.1

Table B.5: We calculate the generalized entropy for $\alpha\epsilon(-1,0,1,2)$ from [4]

$\boldsymbol{\alpha}$	Name of the inequality measure	$\boldsymbol{GE(\alpha)_j}$.
-1	-	$GE(-1)_j = \frac{1}{2}\left[\sum_i^{n_j} \frac{p_i}{P_j}[\hat{y}_i^{-1} - 1]\right]$
0	Population-Weighted Mean Logarithmic Deviation	$MLD_j = GE(0)_j = -\sum_i^{n_j} \frac{p_i}{P_j} log\left(\frac{\hat{y}_i}{\hat{\bar{y}}_j}\right)$
1	Population-Weighted Theil index	$THEIL_j = GE(1)_j = \sum_i^{n_j} \frac{p_i}{P_j}\frac{\hat{y}_i}{\hat{\bar{y}}_j} log\left(\frac{\hat{y}_i}{\hat{\bar{y}}_j}\right)$
2	Population-Weighted Coefficient of Variation	$WCV_j = \sqrt{2 \cdot GE(2)_j} = \sqrt{\sum_i^{n_j} \frac{p_i}{P_j}[\hat{y}_i^{\,2} - 1]}$

Table B.6: Correlation between inequality measures in the cross-section; 180 countries; mean of period 1992–2012

	$GINIW_j$	$GEW(-1)_j$	$MLDW_j$	$THEILW_j$	CVW_j
$GINIW_j$	1.000				
$GEW(-1)_j$	0.946	1.000			
$MLDW_j$	0.943	0.996	1.000		
$THEILW_j$	0.930	0.985	0.996	1.000	
CVW_j	0.971	0.960	0.970	0.971	1.000

Table B.7: Determinants of sigma convergence, panel fixed effects regressions, 5-year averages

	Dependent variable: weighted coefficient of variation (CVW)					
	(1)	(2)	(3)	(4)	(5)	(6)
log(GDP p.c.)	0.513***	0.210	0.166	0.503***	0.207	0.305**
	(0.128)	(0.136)	(0.149)	(0.086)	(0.206)	(0.139)
log(GDP p.c.)^2	-0.056***	-0.024	-0.018	-0.056***	-0.019	-0.033**
	(0.015)	(0.015)	(0.017)	(0.011)	(0.026)	(0.016)
log(GDP p.c.)^3	0.002***	0.001	0.001	0.002***	0.000	0.001*
	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)
Resources rents / GDP	0.028**					
	(0.012)					
Arable land (%)	-0.137***					
	(0.032)					
Trade/GDP		0.012**				
		(0.005)				
FDI/GDP		0.019				
		(0.014)				
Price gasoline			0.001			
			(0.003)			
Price gasoline X Area			0.011**			
			(0.005)			
Bureaucratic quality				-0.002		
				(0.001)		
Polity2				-0.000		
				(0.003)		
log(GDP p.c.) X Federal				-0.991***		
				(0.191)		
Aid/GDP					0.025*	
					(0.013)	
School enrollment secondary					-0.028**	
					(0.013)	
Ethnic inequality						0.017*
						(0.008)
Constant	-1.332***	-0.445	-0.347	-0.862***	-0.519	-0.774*
	(0.366)	(0.398)	(0.441)	(0.205)	(0.546)	(0.403)
# Observations	857	817	676	485	591	841
# Country	176	172	162	126	148	173
R-squared-w	0.318	0.317	0.253	0.322	0.315	0.309
R-squared-b	0.226	0.176	0.203	0.002	0.166	0.216
Country FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses*** p<0.01, ** p<0.05, * p<0.1

Table B.8: Determinants of sigma convergence, panel fixed effects regressions, 5-year averages

	Dependent variable: Theil index $WTHEIL_j$)					
	(1)	(2)	(3)	(4)	(5)	(6)
log(GDP p.c.)	0.086***	0.032	0.029	0.075***	0.022	0.037*
	(0.019)	(0.023)	(0.022)	(0.014)	(0.045)	(0.020)
log(GDP p.c.)^2	-0.010***	-0.004	-0.003	-0.008***	-0.002	-0.004*
	(0.002)	(0.003)	(0.002)	(0.002)	(0.006)	(0.002)
log(GDP p.c.)^3	0.000***	0.000	0.000	0.000***	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Resources rents / GDP	0.005**					
	(0.002)					
Arable land (%)	-0.027***					
	(0.007)					
Trade/GDP		0.002**				
		(0.001)				
FDI/GDP		0.004				
		(0.003)				
Price gasoline			0.000			
			(0.001)			
Price gasoline X Area			0.002***			
			(0.001)			
Bureaucratic quality				-0.000		
				(0.000)		
Polity2				0.000		
				(0.001)		
log(GDP p.c.) X Federal				-0.125***		
				(0.035)		
Aid/GDP					0.004**	
					(0.002)	
School enrollment secondary					-0.004*	
					(0.002)	
Ethnic inequality						0.008***
						(0.003)
Constant	-0.227***	-0.071	-0.068	-0.145***	-0.061	-0.096*
	(0.055)	(0.067)	(0.064)	(0.035)	(0.117)	(0.058)
# Observations	857	817	676	485	591	845
# Country	176	172	162	126	148	173
R-squared-w	0.283	0.284	0.186	0.249	0.266	0.300
R-squared-b	0.213	0.170	0.235	0.005	0.126	0.202
Country FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses*** p<0.01, ** p<0.05, * p<0.1

Table B.9: Determinants of sigma convergence, panel fixed effects regressions, 5-year averages

	Dependent variable: mean log deviation ($WMLD_j$)					
		(2)	(3)	(4)	(5)	(6)
log(GDP p.c.)	0.082***	0.033	0.031	0.070***	0.028	0.019
	(0.018)	(0.021)	(0.020)	(0.014)	(0.040)	(0.017)
log(GDP p.c.)^2	-0.009***	-0.004	-0.004	-0.008***	-0.002	-0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.005)	(0.002)
log(GDP p.c.)^3	0.000***	0.000	0.000	0.000***	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Resources rents / GDP	0.005**					
	(0.002)					
Arable land (%)	-0.024***					
	(0.006)					
Trade/GDP		0.002**				
		(0.001)				
FDI/GDP		0.004				
		(0.003)				
Price gasoline			0.000			
			(0.000)			
Price gasoline X Area			0.002***			
			(0.001)			
Bureaucratic quality				-0.000		
				(0.000)		
Polity2				0.000		
				(0.001)		
log(GDP p.c.) X Federal				-0.101***		
				(0.037)		
Aid/GDP					0.004**	
					(0.002)	
School enrollment secondary					-0.003	
					(0.002)	
Ethnic inequality						0.009***
						(0.002)
Constant	-0.218***	-0.077	-0.076	-0.143***	-0.077	-0.050
	(0.050)	(0.061)	(0.059)	(0.036)	(0.105)	(0.048)
# Observations	857	817	676	485	591	845
# Country	176	172	162	126	148	173
R-squared-w	0.273	0.274	0.176	0.245	0.254	0.434
R-squared-b	0.215	0.156	0.215	0.006	0.117	0.203
Country FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses*** p<0.01, ** p<0.05, * p<0.1

Table B.10: Determinants of sigma convergence, panel fixed effects regressions, 5-year averages

	Dependent variable: $GEW(-1)_j$					
		(2)	(3)	(4)	(5)	(6)
log(GDP p.c.)	0.079***	0.033	0.033*	0.066***	0.032	0.036**
	(0.017)	(0.020)	(0.020)	(0.015)	(0.037)	(0.018)
log(GDP p.c.)^2	-0.009***	-0.004	-0.004	-0.007***	-0.003	-0.004*
	(0.002)	(0.002)	(0.002)	(0.002)	(0.005)	(0.002)
log(GDP p.c.)^3	0.000***	0.000	0.000	0.000***	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Resources rents / GDP	0.005**					
	(0.002)					
Arable land (%)	-0.021***					
	(0.005)					
Trade/GDP		0.002**				
		(0.001)				
FDI/GDP		0.004				
		(0.003)				
Price gasoline			0.000			
			(0.000)			
Price gasoline X Area			0.002**			
			(0.001)			
Bureaucratic quality				-0.000		
				(0.000)		
Polity2				0.000		
				(0.001)		
log(GDP p.c.) X Federal				-0.083**		
				(0.039)		
Aid/GDP					0.005***	
					(0.002)	
School enrollment secondary					-0.003	
					(0.002)	
Ethnic inequality						0.007***
						(0.002)
Constant	-0.213***	-0.081	-0.082	-0.141***	-0.090	-0.098**
	(0.048)	(0.058)	(0.058)	(0.037)	(0.096)	(0.050)
# Observations	857	817	676	485	591	845
# Country	176	172	162	126	148	173
R-squared-w	0.259	0.258	0.165	0.236	0.237	0.277
R-squared-b	0.213	0.140	0.195	0.007	0.105	0.172
Country FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses*** p<0.01, ** p<0.05, * p<0.1

Figure B.1: Predicted regional GDP p.c. in Sub-Saharan Africa

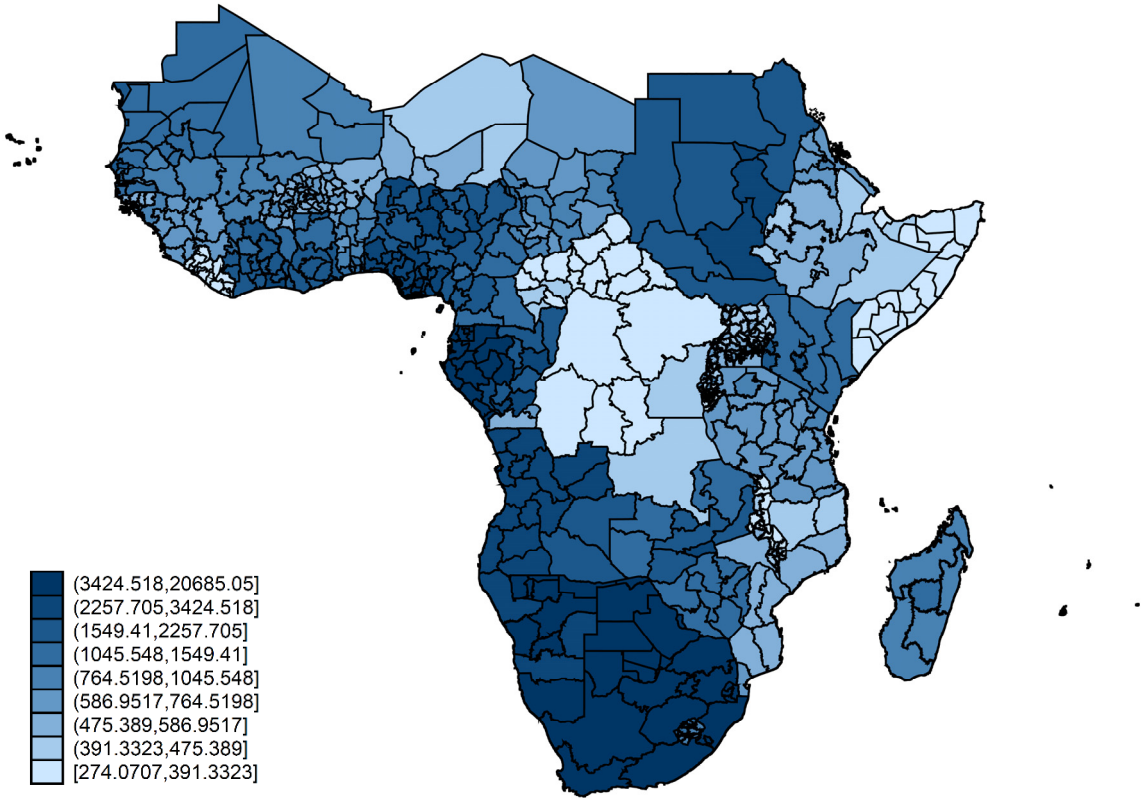
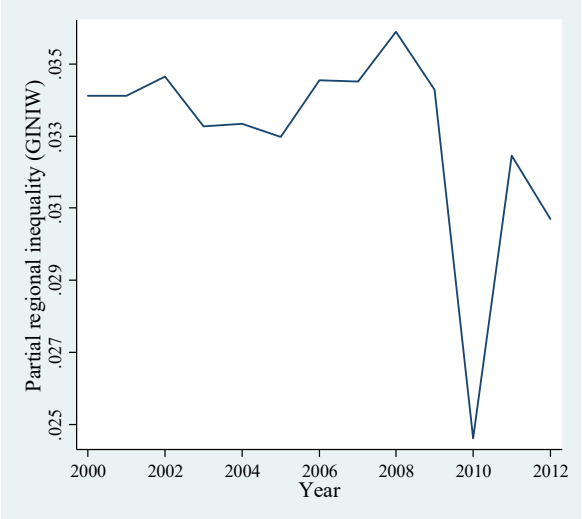


Figure B.3: Predicted regional inequality (GINIW) in Germany 2000-2012



APPENDIX C. SUPPLEMENTARY DATA (ONLINE)

		Regional Inequality (level)										Sigma-Convergence Change in %					Unambiguous Trend all measures
		GINIW		COVW		THEILW		MLDW		GEW(-1)							
		1992-2001	2002-2011	1992-2001	2002-2011	1992-2001	2002-2011	1992-2001	2002-2011	1992-2001	2002-2011	GINIW	COVW	THEILW	MLDW	GEW(-1)	
Country	# Region	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
East Asia & Pacific																	
Australia	11	0.058	0.057	0.109	0.108	0.006	0.006	0.006	0.006	0.006	0.006	-0.3	-0.5	-1.0	-1.0	-1.1	yes
Cambodia	24	0.109	0.115	0.259	0.245	0.029	0.027	0.025	0.024	0.023	0.023	5.1	-5.5	-7.2	-4.5	-1.7	no
China	32	0.070	0.064	0.125	0.115	0.008	0.007	0.008	0.007	0.008	0.007	-8.3	-8.2	-15.7	-15.8	-16.0	yes
Hong Kong	18	0.011	0.013	0.023	0.026	0.000	0.000	0.000	0.000	0.000	0.000	13.3	11.2	24.5	24.4	24.3	yes
Indonesia	33	0.088	0.084	0.169	0.162	0.015	0.013	0.015	0.014	0.016	0.015	-4.0	-4.4	-7.7	-7.0	-6.6	yes
Japan	47	0.052	0.052	0.091	0.091	0.004	0.004	0.004	0.004	0.004	0.004	0.1	0.1	0.3	0.3	0.4	yes
Laos	18	0.081	0.086	0.176	0.175	0.014	0.014	0.013	0.014	0.012	0.013	5.8	-0.5	1.7	4.3	7.1	no
Malaysia	13	0.086	0.088	0.153	0.156	0.012	0.012	0.012	0.013	0.012	0.013	1.6	1.8	4.3	5.0	5.7	yes
Micronesia	4	0.012	0.020	0.023	0.039	0.000	0.001	0.000	0.001	0.000	0.001	67.7	67.8	169.9	168.6	167.4	yes
Mongolia	22	0.142	0.154	0.283	0.297	0.038	0.043	0.037	0.042	0.036	0.042	8.0	4.8	11.9	14.2	16.8	yes
Myanmar	14	0.078	0.078	0.146	0.147	0.010	0.011	0.010	0.010	0.010	0.010	0.4	0.9	3.0	3.1	3.1	yes
New Zealand	16	0.078	0.077	0.142	0.140	0.010	0.010	0.010	0.010	0.010	0.010	-1.9	-2.0	-3.6	-3.4	-3.3	yes
Palau	13	0.108	0.105	0.197	0.208	0.020	0.022	0.021	0.022	0.021	0.022	-3.1	5.6	10.1	5.6	1.6	no
Papua New Guinea	19	0.060	0.065	0.108	0.116	0.006	0.007	0.006	0.007	0.006	0.007	7.4	7.9	15.4	14.6	13.9	yes
Philippines	82	0.124	0.111	0.229	0.203	0.025	0.020	0.025	0.019	0.024	0.019	-10.7	-11.5	-21.3	-21.1	-21.2	yes
Samoa	11	0.079	0.081	0.146	0.151	0.012	0.012	0.012	0.013	0.013	0.014	2.2	3.1	4.5	6.9	9.5	yes
Solomon Islands	8	0.058	0.059	0.114	0.120	0.006	0.007	0.006	0.007	0.006	0.007	1.2	5.7	11.2	10.6	10.0	yes
South Korea	16	0.047	0.041	0.085	0.075	0.004	0.003	0.004	0.003	0.004	0.003	-14.2	-11.4	-21.2	-20.6	-20.0	yes
Taiwan	4	0.016	0.014	0.059	0.052	0.002	0.001	0.002	0.001	0.001	0.001	-11.9	-10.8	-20.1	-19.8	-19.4	yes
Thailand	78	0.097	0.092	0.190	0.176	0.017	0.015	0.016	0.014	0.015	0.014	-5.8	-7.8	-13.2	-12.1	-11.2	yes
Tonga	5	0.041	0.032	0.102	0.080	0.006	0.003	0.007	0.004	0.007	0.004	-22.5	-21.3	-42.7	-43.7	-44.9	yes
Vanuatu	6	0.067	0.067	0.124	0.126	0.008	0.008	0.008	0.009	0.008	0.009	-0.3	2.2	5.0	6.2	7.4	no
Vietnam	8	0.070	0.061	0.127	0.108	0.008	0.006	0.008	0.006	0.008	0.006	-12.9	-14.9	-27.5	-26.7	-26.1	yes
Europe & Central Asia																	
Albania	12	0.052	0.053	0.094	0.096	0.004	0.005	0.004	0.005	0.004	0.005	0.9	1.8	3.2	3.5	3.9	yes
Armenia	12	0.109	0.100	0.198	0.186	0.019	0.017	0.019	0.016	0.019	0.016	-7.5	-5.7	-12.1	-13.4	-14.8	yes

Austria	9	0.055	0.049	0.122	0.109	0.007	0.006	0.007	0.005	0.006	0.005	-10.8	-10.4	-19.0	-18.8	-18.5	yes
Azerbaijan	70	0.094	0.087	0.169	0.160	0.015	0.013	0.015	0.013	0.016	0.013	-7.1	-5.8	-13.2	-15.2	-17.7	yes
Belarus	6	0.023	0.023	0.043	0.044	0.001	0.001	0.001	0.001	0.001	0.001	-1.3	0.6	3.3	3.1	2.9	no
Belgium	15	0.027	0.024	0.050	0.047	0.001	0.001	0.001	0.001	0.001	0.001	-8.3	-7.0	-12.8	-12.9	-13.0	yes
Bosnia and Herzeg.	3	0.014	0.010	0.028	0.026	0.000	0.000	0.000	0.000	0.000	0.000	-28.2	-8.2	-23.2	-25.4	-27.5	yes
Brunei	4	0.081	0.074	0.160	0.148	0.013	0.011	0.014	0.012	0.015	0.013	-9.3	-7.6	-13.9	-13.5	-13.2	yes
Bulgaria	28	0.042	0.042	0.084	0.087	0.003	0.004	0.003	0.004	0.003	0.003	0.9	3.7	6.8	6.2	5.6	yes
Croatia	21	0.037	0.034	0.068	0.061	0.002	0.002	0.002	0.002	0.002	0.002	-7.7	-9.3	-17.5	-18.0	-18.5	yes
Cyprus	6	0.019	0.018	0.036	0.033	0.001	0.001	0.001	0.001	0.001	0.001	-5.9	-7.0	-13.4	-13.9	-14.4	yes
Czech Republic	14	0.034	0.030	0.075	0.070	0.003	0.002	0.003	0.002	0.002	0.002	-10.1	-5.7	-11.1	-11.5	-11.9	yes
Denmark	14	0.053	0.052	0.100	0.098	0.005	0.005	0.005	0.005	0.005	0.005	-2.4	-1.7	-3.4	-3.5	-3.6	yes
Estonia	16	0.044	0.045	0.083	0.084	0.004	0.004	0.004	0.004	0.004	0.004	2.5	1.2	3.0	2.9	2.8	yes
Finland	5	0.035	0.033	0.067	0.062	0.002	0.002	0.002	0.002	0.002	0.002	-6.4	-6.6	-13.3	-13.7	-14.2	yes
France	22	0.040	0.039	0.072	0.070	0.003	0.002	0.003	0.002	0.003	0.002	-3.4	-3.1	-6.0	-5.9	-5.8	yes
Georgia	12	0.095	0.081	0.195	0.176	0.018	0.014	0.016	0.013	0.016	0.013	-13.8	-9.4	-18.1	-18.9	-20.0	yes
Germany	16	0.036	0.033	0.070	0.064	0.002	0.002	0.002	0.002	0.002	0.002	-8.0	-8.6	-16.0	-15.8	-15.6	yes
Greece	14	0.055	0.052	0.105	0.097	0.005	0.005	0.005	0.005	0.005	0.005	-6.6	-7.4	-13.7	-13.1	-12.5	yes
Hungary	20	0.062	0.061	0.132	0.129	0.008	0.008	0.008	0.007	0.007	0.007	-0.9	-2.1	-4.3	-4.1	-4.0	yes
Iceland	8	0.092	0.077	0.185	0.157	0.019	0.013	0.020	0.014	0.021	0.015	-16.7	-15.2	-29.8	-30.6	-31.9	yes
Ireland	26	0.073	0.066	0.138	0.122	0.009	0.007	0.009	0.007	0.009	0.007	-10.5	-11.0	-20.8	-20.7	-20.7	yes
Italy	20	0.021	0.021	0.039	0.039	0.001	0.001	0.001	0.001	0.001	0.001	-0.6	-1.4	-3.2	-3.3	-3.3	yes
Kazakhstan	14	0.026	0.019	0.049	0.035	0.001	0.001	0.001	0.001	0.001	0.001	-24.2	-28.4	-49.7	-50.3	-51.0	yes
Kyrgyzstan	8	0.076	0.069	0.150	0.146	0.011	0.010	0.011	0.009	0.010	0.009	-9.1	-3.1	-8.3	-10.5	-12.7	yes
Latvia	5	0.043	0.044	0.081	0.083	0.003	0.003	0.003	0.003	0.003	0.003	1.3	1.7	3.5	3.2	3.0	yes
Lithuania	10	0.024	0.026	0.044	0.047	0.001	0.001	0.001	0.001	0.001	0.001	8.7	7.7	15.9	15.8	15.8	yes
Luxembourg	4	0.022	0.020	0.048	0.043	0.001	0.001	0.001	0.001	0.001	0.001	-10.0	-11.4	-22.0	-22.3	-22.7	yes
Macedonia	8	0.035	0.037	0.065	0.068	0.002	0.002	0.002	0.002	0.002	0.002	5.7	5.0	9.7	9.8	9.9	yes
Moldova	37	0.058	0.070	0.110	0.135	0.006	0.009	0.006	0.009	0.006	0.008	21.1	22.9	47.0	45.9	45.1	yes
Montenegro	21	0.060	0.064	0.109	0.117	0.006	0.007	0.006	0.007	0.007	0.008	5.9	7.6	16.3	17.0	18.0	yes
Netherlands	14	0.029	0.027	0.055	0.052	0.002	0.001	0.002	0.001	0.002	0.001	-5.9	-4.9	-9.3	-9.1	-9.0	yes
Norway	19	0.074	0.065	0.137	0.119	0.010	0.007	0.010	0.007	0.010	0.007	-11.7	-12.7	-28.1	-31.3	-35.2	yes
Poland	16	0.032	0.026	0.059	0.049	0.002	0.001	0.002	0.001	0.002	0.001	-17.6	-17.0	-30.2	-29.9	-29.5	yes
Portugal	20	0.058	0.052	0.103	0.092	0.005	0.004	0.006	0.004	0.006	0.005	-11.2	-10.6	-20.1	-20.2	-20.4	yes
Romania	41	0.044	0.041	0.088	0.086	0.004	0.004	0.004	0.003	0.004	0.003	-7.7	-2.4	-6.4	-7.1	-7.9	yes
Russia	89	0.086	0.088	0.169	0.174	0.014	0.014	0.013	0.014	0.013	0.014	2.3	2.9	4.9	3.8	2.6	yes
Serbia	25	0.038	0.039	0.069	0.071	0.002	0.002	0.002	0.002	0.002	0.002	1.2	2.2	3.9	3.6	3.3	yes

Slovakia	8	0.021	0.025	0.039	0.047	0.001	0.001	0.001	0.001	0.001	0.001	18.6	21.4	44.8	44.1	43.4	yes
Slovenia	12	0.022	0.021	0.042	0.040	0.001	0.001	0.001	0.001	0.001	0.001	-4.3	-5.5	-10.5	-10.7	-10.9	yes
Spain	18	0.046	0.044	0.081	0.077	0.003	0.003	0.003	0.003	0.003	0.003	-4.6	-4.6	-9.0	-9.0	-9.1	yes
Sweden	21	0.044	0.044	0.079	0.080	0.003	0.003	0.003	0.003	0.003	0.003	1.2	1.0	2.0	2.0	1.9	yes
Switzerland	26	0.048	0.046	0.086	0.082	0.004	0.003	0.004	0.003	0.004	0.003	-4.1	-4.6	-8.6	-8.2	-7.9	yes
Tajikistan	4	0.021	0.018	0.072	0.068	0.003	0.003	0.003	0.003	0.004	0.004	-15.4	-6.4	-12.3	-12.6	-13.1	yes
Turkey	81	0.061	0.059	0.112	0.109	0.006	0.006	0.006	0.006	0.006	0.006	-3.2	-2.7	-5.5	-5.8	-6.2	yes
Turkmenistan	5	0.012	0.014	0.025	0.027	0.000	0.000	0.000	0.000	0.000	0.000	16.4	8.6	7.6	6.4	5.1	yes
Ukraine	27	0.047	0.050	0.096	0.108	0.004	0.005	0.004	0.005	0.004	0.005	5.8	12.9	24.0	21.5	19.3	yes
United Kingdom	4	0.017	0.016	0.047	0.046	0.001	0.001	0.001	0.001	0.001	0.001	-4.7	-3.7	-7.1	-7.1	-7.2	yes
Uruguay	19	0.136	0.130	0.245	0.234	0.030	0.027	0.030	0.027	0.031	0.028	-4.6	-4.6	-9.3	-9.7	-10.1	yes
Uzbekistan	14	0.082	0.081	0.155	0.156	0.012	0.012	0.012	0.012	0.012	0.013	-0.4	0.9	1.9	1.9	2.0	no
Latin America & Caribbean																	
Antigua and Barbuda	8	0.038	0.035	0.076	0.069	0.003	0.002	0.003	0.003	0.003	0.003	-7.9	-8.2	-15.2	-15.2	-15.2	yes
Argentina	24	0.080	0.069	0.183	0.160	0.015	0.012	0.014	0.011	0.013	0.010	-13.6	-12.5	-22.9	-22.6	-22.5	yes
Bahamas	32	0.080	0.067	0.174	0.156	0.017	0.013	0.019	0.015	0.021	0.017	-17.0	-10.5	-19.4	-18.9	-18.6	yes
Barbados	11	0.030	0.027	0.055	0.048	0.001	0.001	0.001	0.001	0.001	0.001	-12.7	-12.3	-22.3	-21.9	-21.5	yes
Belize	6	0.044	0.033	0.080	0.065	0.003	0.002	0.003	0.002	0.003	0.002	-24.8	-18.7	-33.1	-32.3	-31.5	yes
Bermuda	11	0.016	0.018	0.028	0.032	0.000	0.001	0.000	0.001	0.000	0.001	15.7	14.1	30.1	30.1	30.2	yes
Bolivia	9	0.048	0.046	0.094	0.087	0.004	0.004	0.005	0.004	0.005	0.004	-4.2	-7.9	-15.3	-15.5	-15.8	yes
Brazil	27	0.076	0.073	0.135	0.131	0.009	0.009	0.010	0.009	0.010	0.009	-3.5	-3.5	-6.9	-6.9	-7.1	yes
Chile	13	0.072	0.069	0.131	0.125	0.009	0.008	0.009	0.008	0.009	0.009	-4.7	-4.1	-7.9	-7.5	-7.0	yes
Colombia	32	0.061	0.059	0.114	0.111	0.007	0.006	0.007	0.007	0.008	0.007	-2.8	-2.4	-4.9	-5.1	-5.2	yes
Costa Rica	7	0.032	0.027	0.060	0.050	0.002	0.001	0.002	0.001	0.002	0.001	-15.2	-16.5	-32.2	-32.6	-33.0	yes
Cuba	15	0.077	0.070	0.164	0.149	0.012	0.010	0.012	0.010	0.011	0.009	-8.8	-9.0	-16.8	-16.4	-16.1	yes
Dominica	10	0.052	0.043	0.095	0.078	0.005	0.003	0.005	0.003	0.006	0.003	-16.8	-17.2	-38.5	-39.7	-41.1	yes
Dominican Republic	32	0.076	0.072	0.136	0.129	0.009	0.008	0.010	0.009	0.010	0.009	-5.7	-5.2	-10.6	-11.0	-11.4	yes
Ecuador	21	0.041	0.036	0.083	0.074	0.004	0.003	0.004	0.003	0.004	0.003	-13.9	-11.4	-22.3	-22.9	-23.8	yes
El Salvador	14	0.066	0.056	0.118	0.099	0.007	0.005	0.007	0.005	0.007	0.005	-16.1	-16.0	-29.9	-30.1	-30.3	yes
Grenada	7	0.037	0.030	0.071	0.057	0.003	0.002	0.003	0.002	0.003	0.002	-19.1	-20.0	-39.9	-40.6	-41.3	yes
Guatemala	22	0.092	0.081	0.164	0.145	0.013	0.010	0.013	0.011	0.013	0.011	-11.5	-11.8	-21.7	-20.9	-20.1	yes
Guyana	10	0.106	0.108	0.193	0.196	0.019	0.020	0.020	0.021	0.021	0.022	1.5	1.5	3.8	4.6	5.5	yes
Haiti	10	0.076	0.068	0.139	0.124	0.010	0.008	0.010	0.008	0.010	0.008	-11.0	-10.9	-19.1	-18.4	-17.8	yes
Honduras	18	0.079	0.063	0.141	0.116	0.010	0.007	0.010	0.007	0.011	0.007	-19.7	-17.7	-32.2	-31.8	-31.4	yes
Jamaica	14	0.047	0.043	0.086	0.076	0.004	0.003	0.004	0.003	0.004	0.003	-10.1	-11.2	-20.7	-20.2	-19.7	yes
Mexico	32	0.079	0.075	0.145	0.135	0.010	0.009	0.010	0.009	0.010	0.009	-5.7	-6.7	-12.3	-11.9	-11.6	yes

Nicaragua	18	0.105	0.097	0.185	0.171	0.018	0.015	0.018	0.016	0.019	0.016	-8.0	-7.7	-14.6	-14.3	-14.2	yes
Panama	12	0.062	0.057	0.129	0.125	0.009	0.009	0.010	0.010	0.012	0.011	-7.6	-3.2	-5.2	-3.9	-2.4	yes
Paraguay	18	0.132	0.119	0.243	0.216	0.029	0.023	0.028	0.023	0.028	0.023	-10.2	-11.2	-19.9	-18.9	-18.2	yes
Peru	25	0.079	0.073	0.139	0.130	0.010	0.009	0.010	0.009	0.010	0.009	-7.0	-6.7	-12.4	-11.7	-10.9	yes
Puerto Rico	79	0.031	0.030	0.066	0.064	0.002	0.002	0.003	0.003	0.004	0.003	-0.7	-2.6	-6.0	-7.2	-8.4	yes
St. Kitts and Nevis	14	0.029	0.029	0.053	0.052	0.001	0.001	0.001	0.001	0.001	0.001	-2.7	-1.5	-4.7	-5.0	-5.3	yes
St. Lucia	10	0.031	0.026	0.056	0.047	0.002	0.001	0.002	0.001	0.002	0.001	-14.8	-15.7	-29.0	-29.0	-29.1	yes
St. Vincent and G.	6	0.049	0.046	0.093	0.086	0.005	0.004	0.004	0.004	0.004	0.004	-7.3	-7.4	-17.8	-18.3	-18.8	yes
Suriname	10	0.133	0.116	0.246	0.220	0.032	0.026	0.035	0.029	0.040	0.033	-12.8	-10.4	-18.6	-17.4	-16.1	yes
Trinidad/+Tobago	14	0.038	0.029	0.070	0.054	0.002	0.002	0.002	0.002	0.003	0.002	-25.2	-22.2	-37.8	-37.5	-37.4	yes
Venezuela	24	0.083	0.077	0.154	0.143	0.012	0.010	0.012	0.011	0.012	0.011	-7.5	-6.8	-12.8	-12.6	-12.4	yes
Middle East & North Africa																	
Algeria	48	0.084	0.085	0.158	0.158	0.012	0.012	0.012	0.013	0.013	0.013	1.5	0.2	1.1	1.6	2.0	yes
Bahrain	5	0.012	0.008	0.027	0.018	0.000	0.000	0.000	0.000	0.000	0.000	-32.6	-34.6	-46.7	-46.9	-47.0	yes
Djibouti	5	0.101	0.106	0.207	0.220	0.023	0.026	0.025	0.028	0.027	0.031	5.3	6.2	13.7	14.9	16.4	yes
Egypt	26	0.060	0.059	0.113	0.111	0.007	0.006	0.007	0.007	0.008	0.007	-1.8	-2.3	-3.9	-3.3	-2.9	yes
Iran	30	0.055	0.053	0.100	0.095	0.005	0.004	0.005	0.004	0.005	0.005	-4.8	-5.8	-11.4	-11.6	-11.8	yes
Iraq	18	0.079	0.075	0.152	0.142	0.011	0.010	0.010	0.009	0.010	0.009	-5.5	-6.5	-10.9	-9.5	-8.1	yes
Israel	7	0.045	0.042	0.088	0.083	0.004	0.004	0.004	0.004	0.004	0.004	-6.5	-5.5	-10.8	-11.0	-11.2	yes
Jordan	12	0.060	0.058	0.113	0.109	0.006	0.006	0.007	0.006	0.007	0.006	-2.8	-3.0	-6.5	-7.0	-7.7	yes
Libya	32	0.110	0.112	0.194	0.198	0.019	0.020	0.020	0.020	0.021	0.021	1.8	2.0	3.8	3.7	3.6	yes
Morocco	15	0.073	0.069	0.158	0.146	0.012	0.010	0.011	0.009	0.010	0.009	-4.6	-7.9	-13.8	-12.5	-11.3	yes
Oman	8	0.066	0.058	0.117	0.104	0.007	0.005	0.007	0.006	0.007	0.006	-11.8	-11.3	-20.9	-20.4	-19.8	yes
Qatar	9	0.032	0.025	0.069	0.058	0.003	0.002	0.003	0.002	0.003	0.002	-19.8	-15.7	-27.6	-27.9	-28.2	yes
Saudi Arabia	13	0.042	0.042	0.077	0.076	0.003	0.003	0.003	0.003	0.003	0.003	-0.8	-0.9	-2.5	-3.2	-3.8	yes
Syria	14	0.036	0.038	0.068	0.070	0.002	0.002	0.002	0.002	0.002	0.002	5.5	3.7	7.6	7.7	7.7	yes
Tunisia	24	0.076	0.070	0.141	0.127	0.010	0.008	0.009	0.008	0.009	0.008	-8.7	-9.6	-17.5	-16.6	-15.9	yes
U.A. Emirates	8	0.045	0.048	0.085	0.091	0.004	0.004	0.004	0.004	0.004	0.004	5.0	7.5	16.3	17.2	18.2	yes
Yemen	21	0.058	0.048	0.115	0.102	0.007	0.005	0.007	0.006	0.007	0.006	-17.2	-11.1	-19.8	-18.6	-17.4	yes
North America																	
Canada	13	0.031	0.030	0.061	0.059	0.002	0.002	0.002	0.002	0.002	0.002	-3.2	-3.9	-7.8	-8.2	-8.6	yes
United States	51	0.049	0.048	0.090	0.088	0.004	0.004	0.004	0.004	0.004	0.004	-1.6	-1.7	-3.1	-3.1	-3.1	yes
South Asia																	
Afghanistan	32		0.107		0.198		0.019		0.018		0.018						
Bangladesh	6	0.019	0.025	0.036	0.047	0.001	0.001	0.001	0.001	0.001	0.001	33.3	29.3	67.1	66.4	65.6	yes
Bhutan	20	0.076	0.079	0.146	0.142	0.010	0.010	0.010	0.010	0.010	0.011	3.8	-2.7	-1.3	2.8	7.4	no

India	35	0.045	0.046	0.089	0.089	0.004	0.004	0.004	0.004	0.004	0.004	1.1	0.4	0.0	-1.0	-2.0	no
Nepal	5	0.055	0.048	0.103	0.090	0.005	0.004	0.006	0.004	0.006	0.004	-12.8	-12.5	-22.1	-22.1	-22.2	yes
Pakistan	8	0.044	0.045	0.103	0.102	0.006	0.006	0.007	0.006	0.008	0.007	3.4	-1.2	-4.1	-6.1	-8.3	no
Sri Lanka	25	0.072	0.062	0.134	0.114	0.009	0.006	0.009	0.006	0.010	0.007	-13.7	-15.1	-28.6	-29.7	-31.2	yes
Sub-Saharan Africa																	
Angola	18	0.119	0.125	0.283	0.276	0.035	0.034	0.031	0.030	0.028	0.028	5.1	-2.4	-3.8	-2.8	-1.7	no
Benin	12	0.139	0.141	0.268	0.262	0.033	0.032	0.031	0.031	0.030	0.031	1.3	-1.9	-1.7	0.4	2.3	no
Botswana	9	0.101	0.105	0.209	0.211	0.020	0.021	0.019	0.020	0.018	0.019	3.7	1.0	3.0	3.9	4.9	yes
Burkina Faso	44	0.093	0.091	0.197	0.190	0.018	0.016	0.016	0.015	0.015	0.014	-2.0	-3.5	-6.4	-6.0	-5.5	yes
Burundi	17	0.090	0.097	0.215	0.222	0.020	0.021	0.018	0.019	0.016	0.017	6.8	3.0	7.5	8.5	9.5	yes
Cameroon	10	0.055	0.054	0.100	0.098	0.005	0.005	0.005	0.005	0.005	0.005	-2.5	-2.1	-4.4	-4.6	-4.8	yes
Cape Verde	17	0.076	0.054	0.135	0.096	0.010	0.005	0.010	0.005	0.010	0.005	-29.5	-29.2	-52.5	-52.6	-53.1	yes
Central African Rep.	17	0.130	0.125	0.344	0.332	0.050	0.046	0.043	0.040	0.038	0.036	-4.0	-3.7	-6.8	-6.6	-6.5	yes
Chad	18	0.016	0.039	0.059	0.102	0.002	0.005	0.001	0.005	0.001	0.004	137.5	74.0	214.7	223.1	231.2	yes
Comoros	3	0.018	0.015	0.049	0.035	0.001	0.001	0.002	0.001	0.002	0.001	-13.1	-27.1	-45.8	-47.3	-49.0	yes
Côte d'Ivoire	19	0.073	0.073	0.134	0.130	0.009	0.008	0.009	0.009	0.009	0.009	-0.1	-3.1	-5.7	-3.4	-1.0	yes
Dem. Rep. of Congo	9	0.086	0.091	0.209	0.207	0.020	0.019	0.018	0.018	0.016	0.016	6.1	-0.9	-0.8	0.2	1.3	no
Equatorial Guinea	7	0.144	0.107	0.323	0.210	0.046	0.021	0.042	0.020	0.038	0.020	-25.3	-35.0	-53.6	-50.9	-48.5	yes
Eritrea	6	0.077	0.078	0.183	0.173	0.015	0.014	0.014	0.013	0.013	0.012	1.9	-5.7	-9.2	-7.6	-6.0	no
Ethiopia	11	0.060	0.061	0.210	0.197	0.017	0.016	0.014	0.013	0.012	0.012	1.7	-5.9	-9.1	-6.8	-4.3	no
Gabon	9	0.088	0.080	0.160	0.145	0.013	0.011	0.013	0.011	0.014	0.011	-8.8	-8.8	-16.1	-16.0	-16.0	yes
Gambia	6	0.144	0.139	0.289	0.264	0.038	0.034	0.036	0.033	0.034	0.032	-3.1	-8.5	-12.2	-8.8	-5.6	yes
Ghana	10	0.091	0.086	0.170	0.161	0.014	0.013	0.014	0.012	0.014	0.012	-5.4	-5.6	-11.4	-11.8	-12.3	yes
Guinea	8	0.062	0.055	0.175	0.150	0.013	0.010	0.011	0.008	0.010	0.007	-11.2	-14.0	-25.2	-24.4	-23.7	yes
Guinea-Bissau	9	0.134	0.137	0.313	0.330	0.043	0.047	0.038	0.042	0.035	0.039	1.7	5.3	10.4	10.1	9.8	yes
Kenya	8	0.105	0.107	0.219	0.226	0.022	0.023	0.020	0.021	0.019	0.020	1.6	2.9	5.6	5.2	4.8	yes
Lesotho	10	0.075	0.069	0.136	0.124	0.010	0.008	0.010	0.008	0.010	0.008	-8.3	-8.9	-17.2	-17.8	-18.4	yes
Liberia	15	0.057	0.092	0.147	0.193	0.010	0.017	0.009	0.016	0.009	0.016	61.4	31.0	71.0	74.0	77.4	yes
Madagascar	6	0.062	0.065	0.120	0.124	0.007	0.008	0.007	0.007	0.007	0.007	5.2	3.4	6.8	7.3	7.9	yes
Malawi	26	0.089	0.081	0.161	0.150	0.013	0.011	0.012	0.011	0.012	0.010	-8.6	-6.6	-13.6	-14.4	-15.4	yes
Mali	9	0.026	0.035	0.058	0.074	0.002	0.003	0.002	0.003	0.001	0.003	38.6	28.4	69.6	77.1	84.0	yes
Mauritania	13	0.132	0.142	0.303	0.317	0.042	0.045	0.038	0.042	0.036	0.039	7.1	4.7	9.1	8.9	8.9	yes
Mauritius	12	0.035	0.031	0.063	0.056	0.002	0.002	0.002	0.002	0.002	0.002	-10.3	-11.0	-20.6	-20.4	-20.3	yes
Mozambique	10	0.062	0.068	0.128	0.139	0.008	0.009	0.007	0.009	0.007	0.008	9.3	8.9	18.9	19.5	20.2	yes
Namibia	13	0.064	0.066	0.117	0.117	0.007	0.007	0.007	0.007	0.007	0.007	3.2	0.2	1.1	2.0	3.0	yes
Niger	8	0.034	0.031	0.092	0.071	0.004	0.002	0.003	0.002	0.003	0.002	-9.8	-22.6	-36.2	-32.7	-29.4	yes

Nigeria	38	0.111	0.104	0.204	0.189	0.020	0.017	0.019	0.017	0.019	0.016	-6.6	-7.5	-13.5	-13.1	-12.8	yes
Republic of Congo	10	0.078	0.098	0.148	0.181	0.011	0.017	0.012	0.017	0.012	0.017	25.4	22.5	45.6	41.8	38.9	yes
Rwanda	10	0.089	0.089	0.165	0.165	0.013	0.013	0.013	0.013	0.013	0.013	-0.6	0.6	0.8	0.5	0.3	no
Sao Tome and P.	2	0.004	0.003	0.018	0.016	0.000	0.000	0.000	0.000	0.000	0.000	-11.4	-11.4	24.7	28.6	32.9	no
Senegal	11	0.145	0.141	0.272	0.263	0.035	0.033	0.034	0.032	0.033	0.032	-2.6	-3.5	-6.1	-5.6	-5.3	yes
Sierra Leone	4	0.094	0.106	0.226	0.241	0.023	0.026	0.021	0.024	0.019	0.022	12.9	6.3	13.4	14.0	14.8	yes
Somalia	18		0.084		0.202		0.018		0.016		0.015						
South Africa	10	0.073	0.072	0.148	0.143	0.010	0.010	0.010	0.010	0.010	0.009	-1.9	-3.0	-5.3	-4.6	-3.9	yes
South Sudan	3		0.023		0.045		0.001		0.001		0.001						
Sudan	6	0.118	0.108	0.224	0.204	0.024	0.020	0.023	0.019	0.022	0.019	-9.0	-9.0	-16.8	-16.8	-16.8	yes
Swaziland	4	0.037	0.032	0.071	0.062	0.003	0.002	0.003	0.002	0.003	0.002	-14.2	-13.2	-24.8	-25.3	-25.9	yes
Tanzania	26	0.094	0.098	0.216	0.222	0.020	0.022	0.018	0.019	0.016	0.018	4.7	2.9	6.5	7.1	7.7	yes
Togo	5	0.089	0.090	0.172	0.177	0.015	0.015	0.015	0.015	0.015	0.015	1.1	3.0	5.2	4.3	3.4	yes
Uganda	58	0.122	0.129	0.274	0.279	0.032	0.034	0.028	0.030	0.026	0.027	5.7	1.7	4.7	5.8	7.0	yes
Zambia	9	0.100	0.094	0.186	0.177	0.017	0.015	0.016	0.015	0.016	0.015	-5.3	-4.7	-9.5	-9.9	-10.5	yes
Zimbabwe	10	0.126	0.130	0.275	0.277	0.034	0.035	0.031	0.032	0.029	0.031	3.9	0.9	2.7	3.6	4.4	yes